Benjamin Williams

Senior Software Engineer | Apple Platforms

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Senior software engineer with 9+ years shipping production iOS apps. Currently building Splats, an Apple-platform 3D capture product with on-device GPU training and Metal rendering.

SKILLS

Languages	Swift • Objective-C • C++ integration • Rust FFI • Python • Go
Frameworks & Patterns	SwiftUI • UIKit • The Composable Architecture • MVVM • Metal • ARKit
Services & Data	WebSockets • APNs • Bonjour • SQLite • Vapor • Supabase/Postgres
Tooling	Xcode • Tuist • Swift Package Manager • Claude Code • Codex
Delivery	GitHub Actions • Xcode Cloud • TestFlight • Fastlane

EXPERIENCE

Independent Product Engineer | Splats May 2026 - Present

- Architected and shipped a public-beta SwiftUI and TCA product across iPhone, iPad, and Mac that captures imagery, solves camera geometry, trains Gaussian splats on device, and renders them in Metal
- Packaged COLMAP (C++) and the BrushKit trainer (Rust) as versioned XCFrameworks behind Swift APIs with progress, cancellation, input validation, and a recorded engine version for every result
- Instrumented every pipeline stage to find where capture-to-scene time went, showing that reconstruction rather than training dominated, and built on-device benchmarks that hold the workload fixed so comparisons stay valid
- Built paired-Mac compute with Bonjour discovery, trusted device pairing, resumable transfer, live progress, and validation of returned scenes
- Established CI/CD with GitHub Actions PR gates, Xcode Cloud and TestFlight device testing, snapshot suites, and tag-triggered signed and notarized macOS releases
- Directed parallel Claude Code and Codex implementation lanes against written contracts, keeping architecture, license boundaries, and quality decisions under human review

Senior iOS Developer | Phillips Auctioneers Apr 2024 - Apr 2026

- Set the technical direction for a ground-up iOS auction app built with SwiftUI, The Composable Architecture, and Tuist; it shipped to the App Store and remains in active release
- Built the WebSocket live-bidding layer, driving countdown and bid state from an NTP-synced clock so a device's local time could not affect the UI or bid submission
- Led the iOS technical practice through architecture reviews, recurring office hours, pair programming with junior engineers, and diagnosis of complex defects escalated from QA and customers
- Introduced Tuist at project start, structuring the app as independent feature modules so incremental builds stayed fast and project-file merge conflicts near zero as the codebase grew
- Designed and maintained Xcode Cloud CI/CD pipelines for build and release validation

Principal Engineer | Ally Financial Nov 2021 - Dec 2023

- Built an internal SwiftUI demo app that let UX partners review and iterate on designs before production implementation
- Integrated analytics and user-action tracking SDKs across a mixed UIKit, SwiftUI, and embedded-web codebase
- Migrated the team's CI/CD pipelines to Bitrise

Senior iOS Developer | TIAA Jul 2017 - Nov 2021

- Built a declarative MVVM interaction layer over UIKit table and collection views, bringing SwiftUI-style composition to a decade-old app that predated SwiftUI
- Moved the app from Carthage to Swift Package Manager, wrapping internal and third-party dependencies, including binary frameworks, in packages
- Packaged and integrated an internal abstraction over the Transmit authentication SDK, isolating application code from a third-party security dependency
- Led the mobile team's move to a managed Mac setup that gave developers the admin access and update cadence iOS work requires

EDUCATION

BS Computer Science | Clemson University May 2017